# CompLing at ArGuard Shared Tasks: Dialect-Aware Sparse Fusion for Arabic Prompt Safety Classification

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## Abstract

We present our approach in this submission for ArGuard 2026 Subtask B1, which classifies Arabic prompts directed at large language models as *Safe* or *Unsafe*. Our system combines four sparse TF-IDF feature representations with dialect indicators and surface-level features. We trained logistic regression and Naive-Bayes SVM (NBSVM) classifiers and combined their predictions using late fusion. The system achieved a Macro-F1 score of 0.94 and an accuracy of 0.96 on the development set. On the test set, our submission achieved 0.79 accuracy, 0.85 macro precision, 0.71 macro recall, and 0.72 Macro-F1.

## 1 Introduction

Large language models can answer questions, generate instructions, and classify sentiments in many areas (Abiola et al., 2025a,b; Adebanji et al., 2022, 2024; Ojo et al., 2020, 2023, 2024). However, users may also submit prompts that request harmful, abusive, illegal, or otherwise unsafe content. For this reason, a safety classifier can be used as an initial safeguard to estimate whether a prompt is safe before it is sent to a generative model. This task is not limited to detecting harmful words. A user may mention violence, harassment, fraud, privacy, or self-harm for legitimate reasons, such as education, prevention, reporting, or seeking help. Therefore, a useful safety classifier must consider the meaning and context of the prompt, rather than relying only on the presence of harmful terms. Arabic makes this task more difficult because it includes many regional dialects, informal spelling, transliteration, code-switching, and rich morphological variation. The same intention can be written in many different forms. Also, the same words related to harm may appear in both safe and unsafe prompts. Previous research has shown that dialectal and script variation can reveal safety weaknesses that may not appear when models are evaluated only on standardized Arabic text (Al Ghanim et al., 2024; Altakrori et al., 2025; Abdu et al., 2025; Zaghouani, 2026).

ArGuard 2026 focuses on harmful-content detection in Arabic memes and prompts directed at large language models (Alam et al., 2026). Task B focuses on harmful textual prompts. In Subtask B1, systems must classify each prompt as either *Safe* or *Unsafe*. Our submission combines several sparse text representations instead of relying only on a large neural language model. We use four TF-IDF feature blocks based on standard and strongly normalized word and character sequences. We also include dialect labels and 27 surface-level features. Logistic regression and NBSVM separately estimate the probability that a prompt is unsafe. We then combine their predictions using a weighted fusion method selected on the development set. For comparison, we also fully fine-tuned ArabicBERT-Large and CAMeLBERT-Mix. These transformer models were trained with both binary safety labels and auxiliary harm-domain objectives. Figure 1 presents the overall system architecture. The main contributions of this work are a dialect-aware sparse representation that combines word features, character features, dialect information, and surface-level features; an ensemble of NBSVM and logistic regression models with fusion weights selected using development Macro-F1; a comparison with fully fine-tuned ArabicBERT-Large and CAMeLBERT-Mix models, which did not improve the final system; and an analysis of the gap between development and test performance.

## 2 Data Preprocessing and Preparation

### 2.1 Task Data

The task was a binary safety-classification problem for Arabic prompts. Each example included an identifier, prompt text, and a language-variety label. The labeled training and development data also



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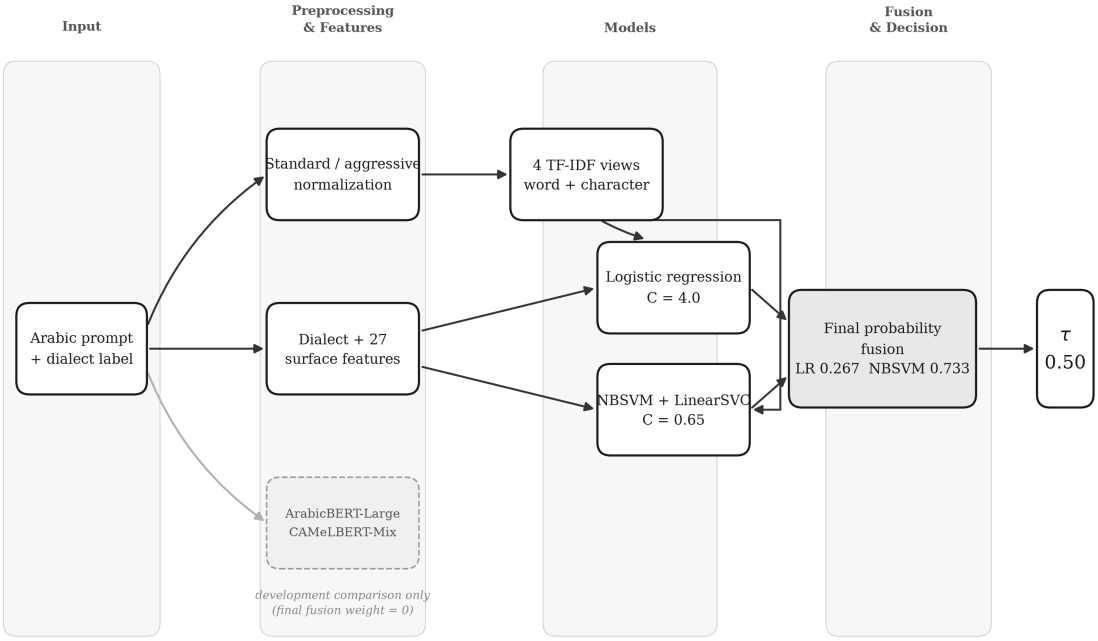



Figure 1: Architecture of the Arabic dialect-aware classification system

| Split | Safe | Unsafe | Total |
|---|---|---|---|
| Model-selection train | 3,393 | 13,377 | 16,770 |
| Development | 475 | 1,930 | 2,405 |
| Test | 2,128 | 3,851 | 5,979 |

Table 1: Dataset composition

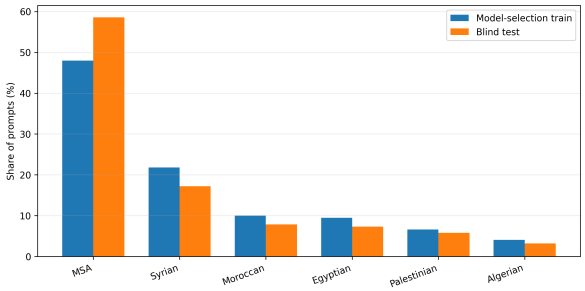



Figure 2: Distribution of Arabic varieties in the model-selection training and test data.

included B1 safety labels and fine-grained harm information. The seven harmful domains were self-harm, harm to others, harassment, adult content, bullying, hate speech, and fraud or illegal activities.

The training data were imbalanced, with unsafe prompts representing 79.8% of the model-selection training set and 80.2% of the development set. The data included six Arabic varieties: MSA, Syrian, Moroccan, Egyptian, Palestinian, and Algerian Arabic. As shown in Figure 2, MSA was the most common named variety in both splits. It represented about 48% of the named-dialect training examples and about 59% of the test examples. The distribution of dialects differed between the training and test data, which may affect how well the model generalizes across Arabic varieties.

## 2.2 Text Normalization

We created two normalized versions of each prompt. The standard version replaced URLs with `URL`, mentions with `USER`, and hashtag symbols with `HASHTAG`. We removed Arabic diacritics and tatweel, mapped alif variants to alif, and mapped alif maqsura to ya. The aggressive version applied additional changes. It simplified hamza-on-waw and hamza-on-ya, mapped ta marbuta to ha, and reduced sequences of three or more repeated characters to two characters. We also added the supplied dialect label to each prompt in the following format:

`dialect: <variety>; text: <prompt>`

This allowed the sparse and transformer models to use dialect information directly.

## 2.3 Surface Features and Resources

We created 27 numeric surface features. These comprised log-scaled raw and normalized character counts, token count, unique-token count, mean token length, Arabic-, Latin-, and digit-character ratios, counts of question marks, excla-

mation marks, quotation marks, line breaks, and repeated-character sequences, binary indicators for URLs, mentions, and hashtags, seven lexical-marker counts, two question indicators, an indicator that aggressive normalization changed the text, and log-scaled dialect-label length. The seven marker inventories contained 25 direct-request expressions, 30 safe-context expressions, 24 violence or self-harm expressions, 18 sexual-content expressions, 23 fraud or illegal-activity expressions, 24 hate or bullying expressions, and 9 negation expressions. They were manually assembled before model fitting from common MSA and dialectal forms relevant to the task taxonomy; for example, the direct-request inventory included variants of "how," "give me," and "I want," while the safe-context inventory included prevention, treatment, reporting, and help-seeking language. Matching was performed on standard-normalized text by summing substring occurrences within each inventory, so a prompt could activate several categories and inflectional or substring matches could produce false positives.

We audited empirical coverage on the held-out development set: at least one inventory matched 2,219 of 2,405 prompts (92.3%). Category-level development coverage was 65.2% for direct requests, 5.4% for safe contexts, 14.8% for violence or self-harm, 20.2% for sexual content, 3.5% for fraud or illegal activity, 4.6% for hate or bullying, and 56.0% for negation; these percentages overlap because inventories were not mutually exclusive. The inventories were not independently annotated by experts and no marker-only ablation was conducted; their validation was therefore limited to this coverage audit and their use within a system selected on held-out Macro-F1. We standardized the 27 numeric features using model-selection training statistics, clipped their standardized values to $[-6, 6]$, and combined them with one-hot dialect indicators plus an unknown-dialect column. For transformer training, we created normalized copies of prompts only when normalization changed the original text. This added 2,787 augmented views during model selection and 3,197 views during final training. During transformer inference, we combined 80% of the original-text probability with 20% of the normalized-text probability.

## 3 System Architecture and Training

### 3.1 Sparse Multi-View Features

The sparse system combined four TF-IDF feature blocks: standard word 1-3 grams with up to 260,000 features, aggressively normalized word 1-2 grams with up to 180,000 features, aggressively normalized character-boundary 2-7 grams with up to 360,000 features, and raw character 2-5 grams with up to 180,000 features. After feature construction, the model-selection matrix contained 340,144 columns. The final model trained on the combined training and development data contained 371,079 columns. To reduce the effect of class imbalance, harm-domain imbalance, and dialect imbalance, we applied sample weights. These weights were clipped to $[0.55, 2.50]$ and normalized to have a mean value of one.

### 3.2 Logistic Regression and NBSVM

We trained a logistic regression classifier with $C = 4.0$, the `liblinear` solver, and a maximum of 2,500 iterations. The classifier estimated the unsafe probability:

$$p_{\mathrm{LR}}(y = \text{unsafe} \mid x).$$

We also trained a Naive-Bayes SVM (NBSVM) classifier (Wang and Manning, 2012). NBSVM first used a Naive Bayes log-count ratio to rescale the TF-IDF features, then trained a linear SVM with $C = 0.65$ and up to 16,000 iterations. We applied a sigmoid function to its decision score to obtain the unsafe probability $p_{\mathrm{NB}}$.

### 3.3 Transformer Models

We also evaluated two fully fine-tuned Arabic transformer models, ArabicBERT-Large and CAMeLBERT-Mix. ArabicBERT-Large used a maximum input length of 192 tokens, at most 3 training epochs, a learning rate of $8 \times 10^{-6}$, a per-device training batch size of 4, a per-device evaluation batch size of 16, and 8 gradient-accumulation steps (effective training batch size 32 on the single-device run). CAMeLBERT-Mix used the same maximum input length, at most 4 epochs, a learning rate of $1.8 \times 10^{-5}$, training and evaluation batch sizes of 16 and 48, respectively, and 2 gradient-accumulation steps (effective training batch size 32). Both used AdamW with weight decay 0.02, cosine decay, an 8% warmup measured in optimizer-update steps, mixed precision, label smoothing of 0.02, and class and example weights. The binary

loss was weighted cross-entropy multiplied per example by $(1-p_t)^{\gamma}$, with $\gamma = 0.60$ for ArabicBERT-Large and $\gamma = 0.75$ for CAMeLBERT-Mix. Both models used a multitask classification head. The primary task predicted the B1 safety label, while an auxiliary task predicted one of the seven harm domains when the label was available:

$$\mathcal{L} = \mathcal{L}_{\mathrm{B1}} + 0.18\,\mathcal{L}_{\mathrm{harm}}. \quad (1)$$

Evaluation and checkpoint saving occurred after every epoch. Early stopping monitored development Macro-F1 with patience 2; the retained checkpoint was the epoch with the highest development Macro-F1, and ties followed the trainer's checkpoint ordering. The decision threshold reported for each retained checkpoint was selected on the same development set by evaluating midpoints between distinct predicted probabilities, retaining thresholds within $10^{-12}$ of the best Macro-F1, and choosing the tied threshold nearest 0.5. Model-selection training used one random seed (42) for each transformer. Consequently, the paper cannot report across-run mean or variance for these development results; ArabicBERT-Large used two seeds (42 and 3407) only in the subsequent train/development refit used to form test probabilities, whereas CAMeLBERT-Mix used seed 42. The unusually low development scores and low selected thresholds should therefore be interpreted as an undiagnosed, configuration-specific negative result, potentially affected by the combined focal loss, weighting, augmentation, auxiliary supervision, or calibration procedure rather than evidence about the general capability of either pretrained encoder. Under this evaluated configuration, both transformer branches received zero weight in the submission.

### 3.4 Fusion

We combined model probabilities using weighted averaging and selected the final configuration on the development set using Macro-F1. We evaluated 979 weight settings and searched for a stable decision threshold. The final submitted model combined logistic regression and NBSVM as follows:

$$p_f = 0.267\,p_{\mathrm{LR}} + 0.733\,p_{\mathrm{NB}}, \qquad \tau = 0.50. \quad (2)$$

After selecting the configuration, we retrained the sparse models using the cleaned combination of training and development data.

| System | Macro-F1 | Accuracy | Threshold |
|---|---|---|---|
| Logistic regression | 0.9321 | 0.9568 | 0.5750 |
| NBSVM | 0.9380 | 0.9609 | 0.5373 |
| ArabicBERT-Large | 0.4950 | 0.6811 | 0.3675 |
| CAMeLBERT-Mix | 0.4999 | 0.7106 | 0.2054 |
| Final sparse fusion | **0.9396** | **0.9626** | 0.5000 |

Table 2: Development-set results

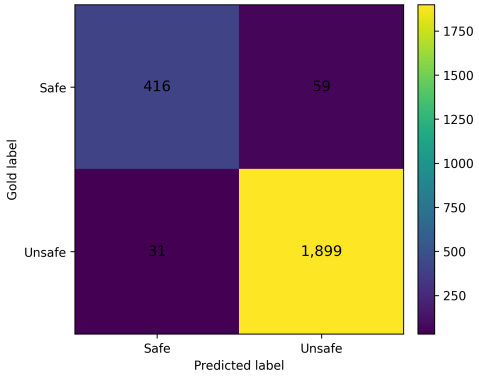



Figure 3: Development confusion matrix for the final sparse fusion system.

## 4 Experimental Results

### 4.1 Development Results

Table 2 shows development-set performance. NBSVM was the strongest individual model, with a Macro-F1 score of 0.9380. Logistic regression achieved 0.9321 Macro-F1. The final sparse fusion achieved the highest score among the configurations evaluated in this study, with a Macro-F1 score of 0.9396 and an accuracy of 0.9626. Under the single-seed transformer configurations described above, ArabicBERT-Large achieved 0.4950 Macro-F1 and CAMeLBERT-Mix achieved 0.4999, so they were excluded from the submission. This comparison establishes only that the selected sparse fusion was stronger than these two particular training runs on this development split. The transformer run-to-run variance was not estimated and the low scores were not diagnosed through controlled loss, weighting, augmentation, auxiliary-task, and learning-rate ablations.

For the final fusion, safe-class precision, recall, and F1 were 0.9306, 0.8758, and 0.9024. Unsafe-class precision, recall, and F1 were 0.9699, 0.9839, and 0.9769. The system correctly classified 416 safe prompts and 1,899 unsafe prompts. It incorrectly classified 59 safe prompts as unsafe and 31 unsafe prompts as safe.

| Metric | Score |
|---|---|
| Accuracy | **0.787** |
| Macro precision | **0.848** |
| Macro recall | **0.708** |
| Macro-F1 | **0.724** |

Table 3: Test result. Macro-F1 was the main task metric.

### 4.2 Test Results

On the test set, the submitted system achieved **0.787 accuracy**, **0.848 macro precision**, **0.708 macro recall**, and **0.724 Macro-F1**. The results show that the system maintained relatively high precision, indicating that many of its predictions were correct. However, the lower macro recall suggests that the model did not identify all examples equally well across the two classes. The test Macro-F1 score was lower than the development score, which suggests that the blind test data differed from the development data.

## 5 Error Analysis

Figure 3 shows the confusion matrix for the final sparse fusion system. The main error was falsely classifying safe prompts as unsafe. Safe-class recall was 0.8758, while unsafe-class recall was 0.9839. The model correctly identified most unsafe prompts, with relatively few unsafe examples predicted as safe. However, a larger number of safe prompts were predicted as unsafe. This suggests that the model was very effective at recognizing unsafe prompts but sometimes rejected safe prompts that contained harmful or sensitive vocabulary.

## 6 Ethical Considerations

Harmful-prompt classification involves sensitive material, including violence, harassment, hate speech, self-harm, sexual content, and illegal activities. We used the ArGuard data only for research and did not reproduce harmful prompts in this paper.

## 7 Limitations

The lexical lists were created manually, used substring matching, and were not tested separately. The transformer models were trained with one random seed, and their weak results were not examined through controlled tests of hyperparameters and training objectives. The system and threshold were selected using the development set, which may have caused overfitting, especially because of the dialect shift and lower Macro-F1 on the blind test set. Therefore, the findings apply only to the tested models and data splits and should not be generalized.

## 8 Conclusion

We presented our submission to ArGuard 2026 Subtask B1, which classified Arabic prompts as *Safe* or *Unsafe*. Our final system combined four word- and character-level representations, dialect indicators, surface-level features, logistic regression, and NBSVM. The final fusion used weights of 0.267 for logistic regression and 0.733 for NBSVM. It achieved a Macro-F1 score of 0.9396 and an accuracy of 0.9626 on the development set. On the test set, the system achieved an accuracy of 0.787, a macro precision of 0.848, a macro recall of 0.708, and a Macro-F1 score of 0.724. These results show that the selected sparse lexical configuration was effective for this submission.

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